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(1) Research Program/Project/Study Title: Pollen Viability of Different Coconut (<i>Cocos nucifera</i> L.) Varieties as Affected by Various Culture Media	
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Program or Project has component (if no proceed to 2)? ☐ Yes ☐ No

Component Title:	
Component Leader:	
Academic Rank:	

(2) Category (please check) ☐ Social Science/Education /Humanities & other related fields ☒ Natural/Biological Sciences ☐ Food Research/ ☐ Developmental/ Computer/Engineering/ Architecture ☐ Extension ☐ Poster		(3) Research Center [Check the concerned center] ☐ SSITRDC Social Science and Industrial Technology Research Development Center ☐ LRDC Livestock Research Development Center ☐ FFRDC Fisheries and Food Research Development Center ☒ CSRDC Crop Science Research Development Center	
(4) Site of implementation			
Municipality/City:		Mambusao	
Province:		Capiz	
Region:		VI	
(5) Classification Research: Development: ☐ Basic ☐ Pilot Testing ☒ Applied ☐ Tech. Promotion/Commercialization		(6) Mode of Implementation ☒ Single Authorship/Proponent ☐ Collaborative Research (within campus) ☐ Collaborative Research (intercampus) ☐ Collaborative Research (external agency)	
(7) Sector/Commodity Coconut		(8) Discipline/Field of Specialization Agriculture	

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(9A) Abstract (This has a maximum of 250 words. It contains 1-2 sentences of rationale, followed by objectives, brief methodology, major findings, and implications of impact of the study.)

The study was conducted at the Crop and Soils Laboratory of Capiz State University, Burias Campus, Mambusao, Capiz. It was laid out using a Completely Randomized Design (CRD) with two factors, where Factor A consisted of different coconut varieties: A1 – Laguna Tall, A2 – Baybay Tall, and A3 – Tagnanan Tall. Factor B involved different culture media: B1 – platelet count agar and B2 – potato dextrose agar. Each treatment was replicated three times. This was conducted to evaluate the pollen viability of different coconut varieties using various culture media.

Results revealed that the viability of pollen was highly significant among the different coconut varieties. However, there was no significant effect on pollen viability when applied with different culture media. Findings also showed that the active germinated pollen varied significantly with the coconut varieties using different culture media. Germinated pollen showed a white to light brown color and spheroidal shape of grain. Active germinated pollen was characterized by the presence of a pollen tube. It was observed that using potato dextrose agar resulted in a longer pollen tube than platelet count agar. Ungerminated pollen illustrated a dark brown to black color and spheroidal shape of grain. The study revealed that the length of spheroidal grain ranged from 31 to 43 μm .

(9B) Keywords (Write 3-6 words you used in the study, preferably not found in the title because words in the title are already given keywords. This would also include process, or methodology employed.)

Pollen viability test, Pollen viability, coconut, coconut pollen, culture media

1. **Introduction** (This includes the rationale, background of the study, and objectives which are not divided by subheadings. Incorporate at least three (3) reviews)

Coconut (*Cocos nucifera* L.) is a crucial agricultural commodity in tropical regions, valued for its versatile products ranging from food and beverages to industrial materials. One of the essential factors for successful coconut cultivation is the viability of its pollen, which directly affects fertilization and fruit set. Pollen viability is influenced by various factors, including the genetic characteristics of the coconut variety and the conditions under which the pollen is cultured.

Pollen viability refers to the capacity of pollen grains to germinate and successfully fertilize an ovule, leading to seed formation. It is a critical determinant of reproductive success in plants and is influenced by both intrinsic and extrinsic factors (Shivanna & Johri, 1985). Intrinsic factors include the genetic makeup and physiological state of the pollen, while extrinsic factors encompass environmental conditions and the composition of the culture media used for pollen germination and storage (Stanley & Linskens, 1974).

The viability of coconut pollen can vary significantly among different varieties due to genetic differences. Studies have shown that pollen from different coconut varieties exhibits varying degrees of viability and germination rates when subjected to similar conditions (Perera et al., 2009). This variation underscores the importance of selecting appropriate culture media tailored to the specific needs of each variety to maximize pollen viability and subsequent fertilization success.

Culture media play a crucial role in maintaining pollen viability by providing the necessary nutrients and optimal conditions for pollen germination. Different culture media compositions, including varying concentrations of sucrose, boric acid, calcium nitrate, and other nutrients, have been evaluated for their effectiveness in supporting coconut pollen

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viability (Heslop-Harrison & Heslop-Harrison, 1985). For instance, research by Jayasekara et al. (2002) demonstrated that a culture medium with 15% sucrose and 0.01% boric acid significantly improved the germination rate of coconut pollen compared to other media compositions.

Environmental factors, such as temperature and humidity, also influence pollen viability and can interact with the culture media to affect germination rates (Sedgley & Griffin, 1989). Optimal temperature and humidity levels must be maintained during the culture process to ensure high pollen viability. Deviations from these optimal conditions can lead to reduced pollen germination and lower fertilization rates (Baker & Baker, 1979).

In the context of coconut cultivation, improving pollen viability through optimized culture media can have significant implications for breeding programs and commercial production. Enhanced pollen viability ensures higher rates of successful fertilization, leading to increased fruit set and improved yield. This is particularly important for hybridization programs aimed at developing new coconut varieties with desirable traits, such as disease resistance, drought tolerance, and high yield potential (Samosir et al., 2016).

Moreover, understanding the interactions between different coconut varieties and culture media can aid in the development of standardized protocols for pollen handling and storage, which are essential for large-scale breeding and propagation efforts (Perera et al., 2009). By identifying the most effective culture media for each variety, researchers and growers can enhance the efficiency and success of coconut breeding programs, ultimately contributing to the sustainability and productivity of coconut agriculture.

The viability of coconut pollen is a critical factor influencing the reproductive success and yield of coconut crops. Different coconut varieties exhibit varying degrees of pollen viability, which can be significantly affected by the composition of the culture media. Optimizing culture media to support high pollen viability is essential for improving fertilization rates, enhancing breeding programs, and ensuring higher yields in coconut cultivation.

Objectives of the Study

Generally, the study aimed to determine the pollen viability of different coconut varieties using various growth media. Specifically, the goal of the study was to:

1. Determine the pollen viability of different coconut varieties;
2. Identify the pollen viability of coconut pollen using various culture media;
3. Evaluate the active germinated pollen using various culture media; and
4. Find out the pollen morphology of coconut using various culture media.

(10) **Materials and Methods or Methodology** (This includes the materials, methods, duration & location of the study. Materials and Methods are headings used in Natural Science researches and methodology is Social Sciences. Presentation of materials may not necessarily be in separate heading but may be incorporated in the specific methods used. Justify paragraphs with main headings in caps.)

The study was conducted at the Crop and Soils Laboratory of Capiz State University, Burias Campus, Mambusao, Capiz. It was laid out using a Completely Randomized Design (CRD) with two factors, where Factor A consisted of different coconut varieties: A1 – Laguna Tall, A2 – Baybay Tall, and A3 – Tagnanan Tall. Factor B involved different culture media: B1 – platelet count agar and B2 – potato dextrose agar. Each treatment was replicated three times. Differences among treatment means were compared using the Least Significant Difference (LSD) test, interpreted at a 5% level of significance. The study used the ImageJ application to determine pollen viability (Figures 6 and 7). Data gathered were analyzed using the Statistical

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Tool for Agricultural Research (STAR) and were interpreted at a 5% level of significance.

Cultural Practices and Management

Preparation of coconut pollen

Pollen used in the study was procured from PCA Aroman, North Cotabato. The pollen included Laguna Tall (A1), Baybay Tall (A2), and Tagnanan Tall (A3).

Preparation of culture media

The culture media were prepared by heating 100 mL of distilled water. Then, 0.5 g of agar, 10 g of sucrose, and 1 mL of boric acid solution were added while stirring continuously until the materials were completely dissolved.

Pollen Viability Test

Pollen viability tests were conducted by preparing culture media in which the pollen was cultivated. The agars prepared in the study included platelet count agar (B1) and potato dextrose agar (B2).

Pollen viability test were determined by:

$$\% \text{ Pollen Germination} = \frac{\text{Number of germinated pollen}}{\text{Total number of pollen counted}} \times 100$$

Active germinated pollen were determined by:

$$\% \text{Active pollen} = \frac{\text{Total number of pollen tube counted}}{\text{Number of germinated pollen}} \times 100$$

(11) **Results and Discussions** (This part contains presentation of results (data) with reference to the table and figure, to be followed by discussion, with support of reviewed literature. Sequence shall be based on the objectives of the study. Tables are formatted without the vertical lines. Horizontal lines are set at the top and bottom portions as well as below table headings and above major subdivisions. Justify paragraphs with main headings in caps.)

Pollen viability test result

Data on the pollen viability (%) of different coconut varieties using various culture media are presented in Table 1. Analysis of Variance (ANOVA) indicated that the results were highly significant among the different coconut varieties but not significant on the effect of culture media. This implied that the viability of pollen varied significantly with the different coconut varieties. However, there was no significant effect of different culture media on pollen viability. The Least Significant Difference (LSD) test revealed that Tagnanan Tall (53.54%) recorded the highest pollen viability, which was significantly higher than Baybay Tall (50.16%), and comparable to Laguna Tall (47.27%). The viability of pollen using potato dextrose agar (51.19%) was comparable to that using platelet count agar (49.45%). With regard to culture media, means ranged from 49.45% to 51.19%.

Data on the active germinated pollen (%) of different coconut varieties using various culture media are presented in Table 2. ANOVA indicated that the results were highly significant for both the effect of coconut varieties and culture media. The LSD test revealed

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that Tagnanan Tall (11.45%) had the highest active germinated pollen, which was comparable to Baybay Tall (10.04%). Laguna Tall showed a significantly lower active germinated pollen (7.67%). The results further revealed that potato dextrose agar (12.57%) yielded a significantly higher result than platelet count agar (6.86%).

This concurs with the study by Perera et al. (2009), which indicated that the viability of coconut pollen can vary significantly among different varieties due to genetic differences. Studies have shown that pollen from different coconut varieties exhibits varying degrees of viability and germination rates under similar conditions.

Coconut pollen morphology observation

The pollen extracted from the male flowers of Laguna Tall, Baybay Tall, and Tagnanan Tall were powdery and yellowish in color (Figure 1). Under a trinocular compound microscope at 4X magnification, some morphological characteristics of coconut pollen were observed (Figures 2 and 3). Germinated pollen showed a white to light brown color and a spheroidal shape of grain. Active germinated pollen was characterized by the presence of a pollen tube (Figures 4 and 5). It was observed that using potato dextrose agar resulted in longer pollen tubes compared to platelet count agar. Ungerminated pollen illustrated a dark brown to black color and a spheroidal shape of grain (Figures 6 and 7). Figures 8 and 9 revealed that the length of the spheroidal grain ranged from 31 to 43 μm .

(13) Conclusions and Recommendations (This part shall be written in narrative form and arranged based on the objectives and findings. Conclusions are general statements answering the objectives. Recommendations are based on the conclusions. They may be split or combined based on their length, which means, conclusions and recommendations may be split under separate headings if they are long. Justify paragraphs with main headings in caps.)

Conclusions

Based on the findings of the study, the following conclusions have been reached:

1. The viability of pollen varied significantly among the coconut varieties used in the study.
2. The pollen viability was not significantly influenced by the different culture media used.
3. The application of various culture media had a significant effect on the rate of active germinated pollen in coconut.
4. Potato dextrose agar resulted in a longer and more dominant pollen tube presence compared to platelet count agar.

Recommendations

Based on the result of this study, the following recommendations are forwarded:

1. Use potato dextrose agar when preparing culture media to ensure more reliable results for pollen viability.
2. Conduct similar studies with enhanced culture media ratios to improve seed viability test results.

(14) Output (Indicate the specific products, processes or services which the project has produced; quantify when possible.)

(15) Technology Transfer Activity Conducted (Indicate the specific products, activity design, processes or services transferred as extension activities)

(16) Economic Analysis (This should present the direct cost of project implementation, expenses in the production of technology, revenues generated from technology produced, return on investment (ROI) and payback

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period.

(17) Acknowledgments (Include only important persons and institutions with involvement in funding and conduct of the research. Justify paragraphs with main headings in caps.)

(18) References (It should include only works mentioned in the text and should be arranged alphabetically. Citations shall not be less than 13 entries. This must follow the APA latest edition format, in hanging indention with 12-point space in between.)

- Baker, H. G., & Baker, I. (1979). Starch in angiosperm pollen grains and its evolutionary significance. *American Journal of Botany*, 66(5), 591-600.
- Heslop-Harrison, J., & Heslop-Harrison, Y. (1985). Germination of pollens on nutrient media: Pollen-stigma interaction and subsequent fertilization. *Journal of Cell Science*, 2, 1-15.
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APPENDICES

Table 1. Pollen viability (%) of different coconut varieties using various culture media.

Viability of Pollen (%)				
Treatment	Culture Media ^{ns}			
Varieties**	B1 – Platelet Count Agar	B2 – Potato Dextrose Agar	Total	Mean
A1 – Laguna Tall	46.20	48.33	94.53	47.27 ^b
A2 – Baybay Tall	49.76	50.55	100.31	50.16 ^b
A3 – Tagnanan Tall	52.39	54.70	107.09	53.54 ^a
Total	148.35	153.58		
Mean	49.45	51.19		

Table 2. Active Germinated Pollen (%) of different coconut varieties using various culture media.

Active Germinated Pollen (%)				
Treatment	Culture Media**			
Varieties**	B1 – Platelet Count Agar	B2 – Potato Dextrose Agar	Total	Mean
A1 – Laguna Tall	6.27	9.07	15.34	7.67 ^b
A2 – Baybay Tall	6.15	13.92	20.07	10.04 ^a
A3 – Tagnanan Tall	8.17	14.74	22.91	11.45 ^a
Total	20.59	37.73		
Mean	6.86 ^b	12.57 ^a		



Figure 1. Coconut pollen sample extracted from male flower of Baybay tall procured from PCA Aroman, North Cotabato.

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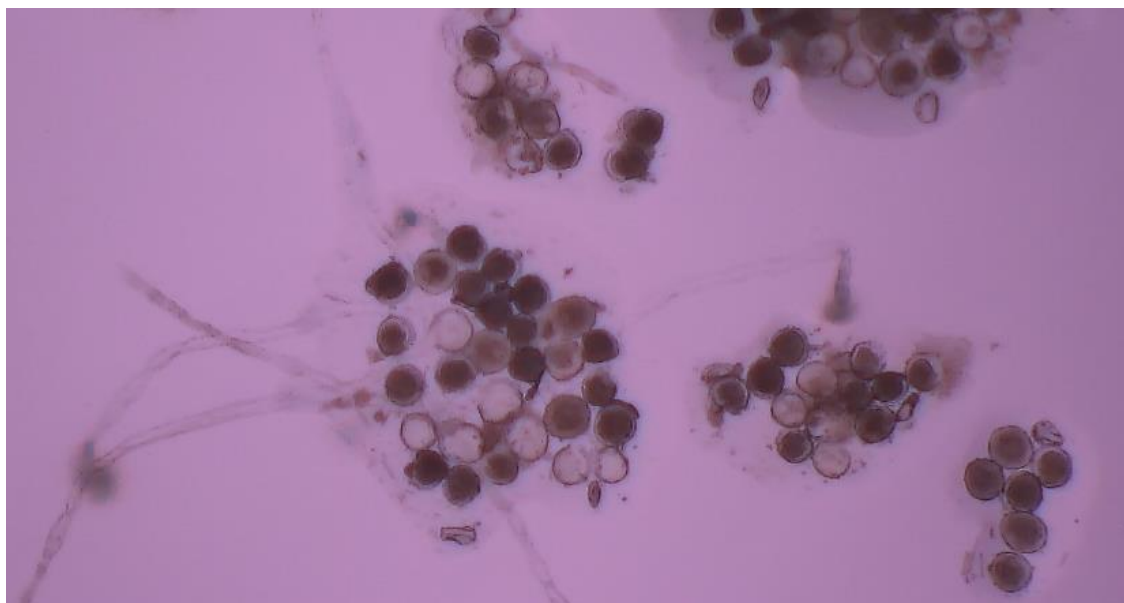


Figure 2. Germinated and ungerminated pollen of a Laguna Tall using platelet count agar as a culture media (A1B1) in a trinocular compound microscope at 4X magnification.

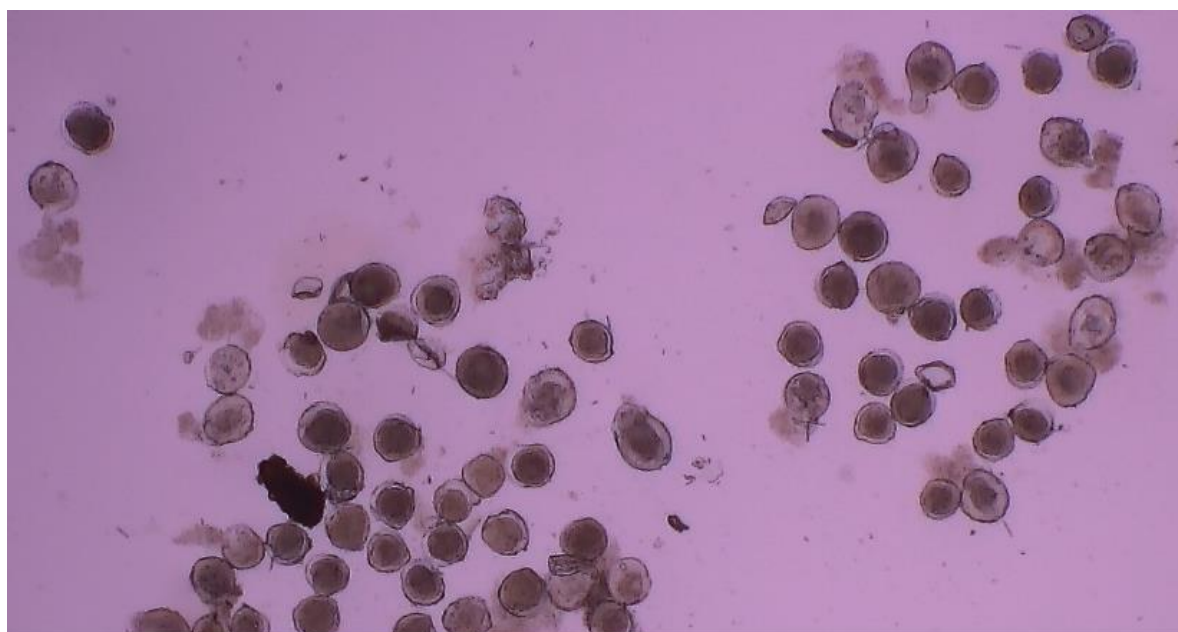


Figure 3. Germinated and ungerminated pollen of a Laguna Tall using potato dextrose agar as a culture media (A1B2) in a trinocular compound microscope at 4X magnification.

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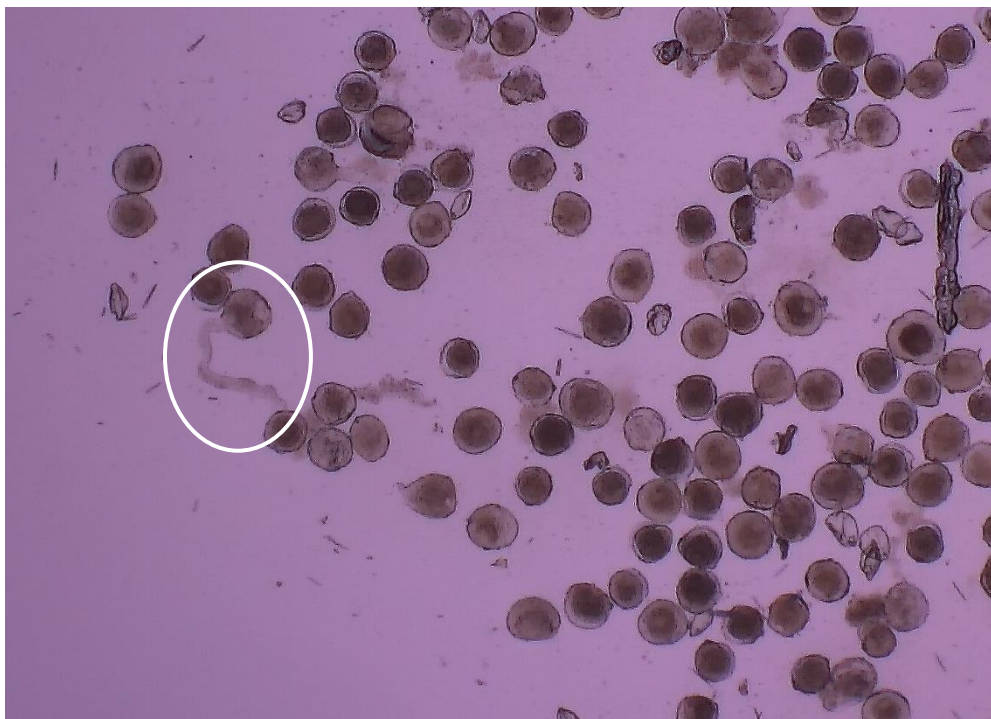


Figure 4. Active germinated pollen of Baybay Tall using platelet count agar (A2B1).



Figure 5. Active germinated pollen of Baybay Tall using potato dextrose agar (A2B2).

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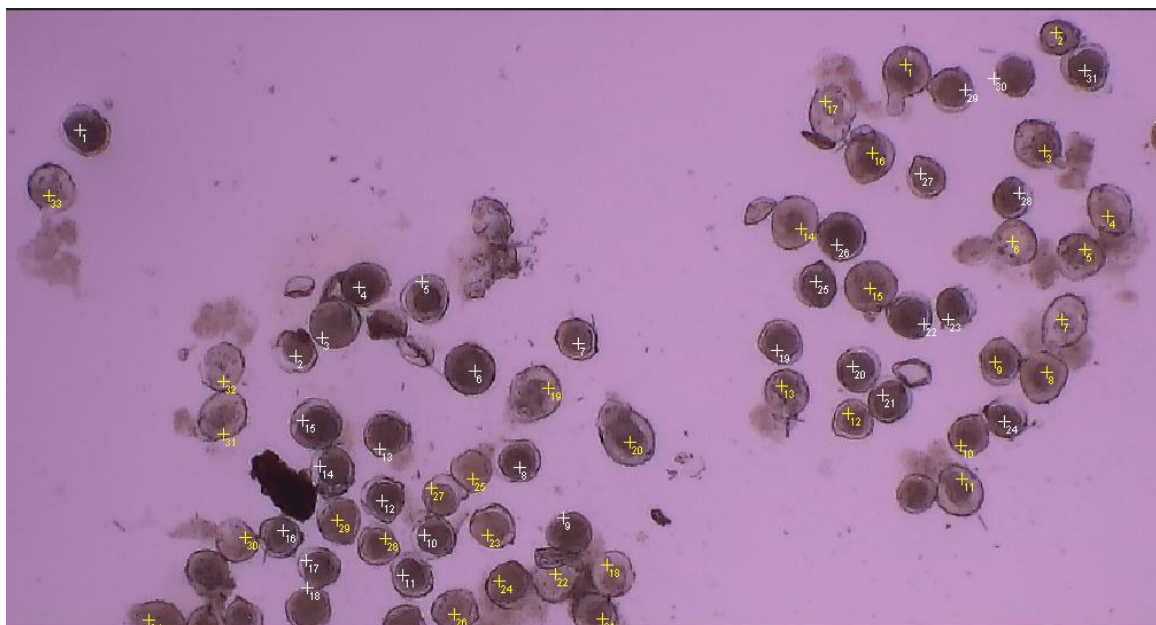


Figure 6. Counting of pollen using image J application: Yellow cross pointed tool indicates germinated pollen; white cross pointed tool indicates ungerminated pollen.

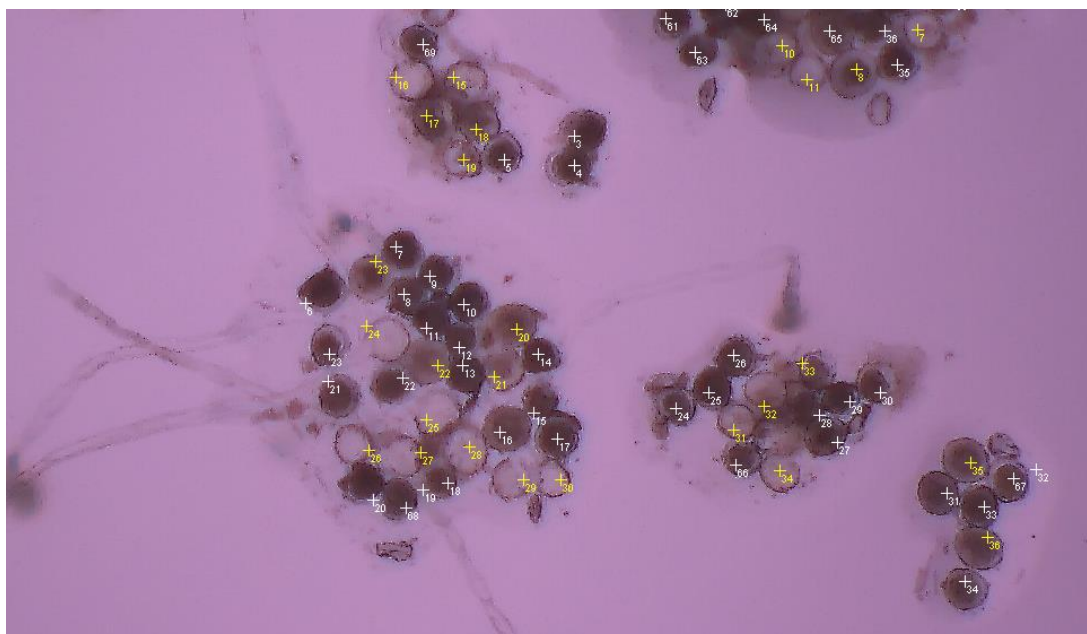


Figure 7. Counting of pollen using image J application: Yellow cross pointed tool indicates germinated pollen; white cross pointed tool indicates ungerminated pollen.

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Figure 8. Measurements of Laguna Tall pollen (largest – C1 and smallest – C2) using potato dextrose agar (A1B1).

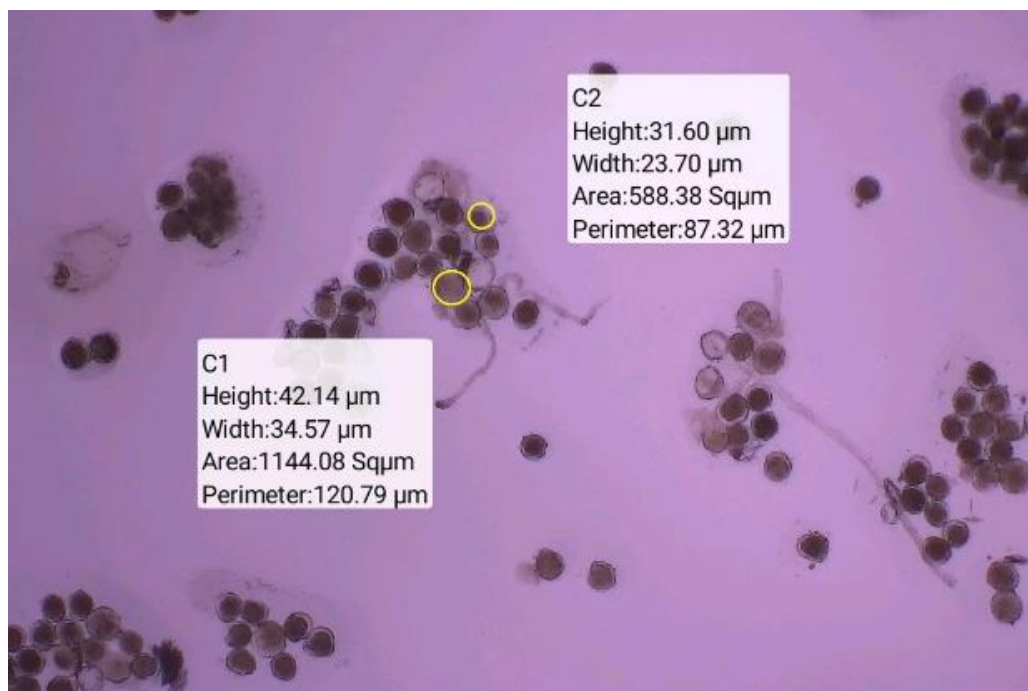


Figure 9. Measurements of Laguna Tall pollen (largest – C1 and smallest – C2) using platelet count agar (A1B2).